

Sparse and Deep Gaussian Processes Closure for 2-D Fluids and Ocean Flows

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Motivation and Challenges

Motivation:

Fully resolved physics:

- High accuracy at High Cost
- Need for reduction

Under-resolved physics:

- Lower accuracy at Lower Cost
- Truncation/discretization error

Challenges:

- Complex nonlinear, multiscale fluid dynamics with unresolved sub-grid physics
- Truncated numerical models introduce truncation and discretization biases
- Regime variability (Reynolds number shifts, Transient forcing) and evolving flow structures

Needs:

- Fast, online data-driven correction of LR dynamics embedded directly into the solver
- Uncertainty-aware predictions of resolved fields and wake metrics
- Transferable across resolutions, domains, and flow conditions

Sparse variational Gaussian Process (SVGP) Closure

Why Gaussian Process?

- Structured priors:** physical smoothness and invariances encoded through Kernel
- Non-parametric flexibility:** expressive function modeling without rigid parametric assumptions
- Uncertainty quantification:** posterior variance enables OOD detection, confidence-based gating of predictions, and natural extension to stochastic solvers

Gaussian Process Regression

$$f(x) \sim \mathcal{GP}(\mu(x), k(x, x'))$$

$$\mu_*(x_*) = k_{*X} K_{XX}^{-1} y, \quad \sigma_*^2(x_*) = k_{**} - k_{*X} K_{XX}^{-1} k_{X*}$$

Multi-Output GP (Output-as-Input view)

$$f(x, i) \sim \mathcal{GP}(\mu(x, i), k((x, i), (x', j))), \quad i \in \{1, \dots, D\}.$$

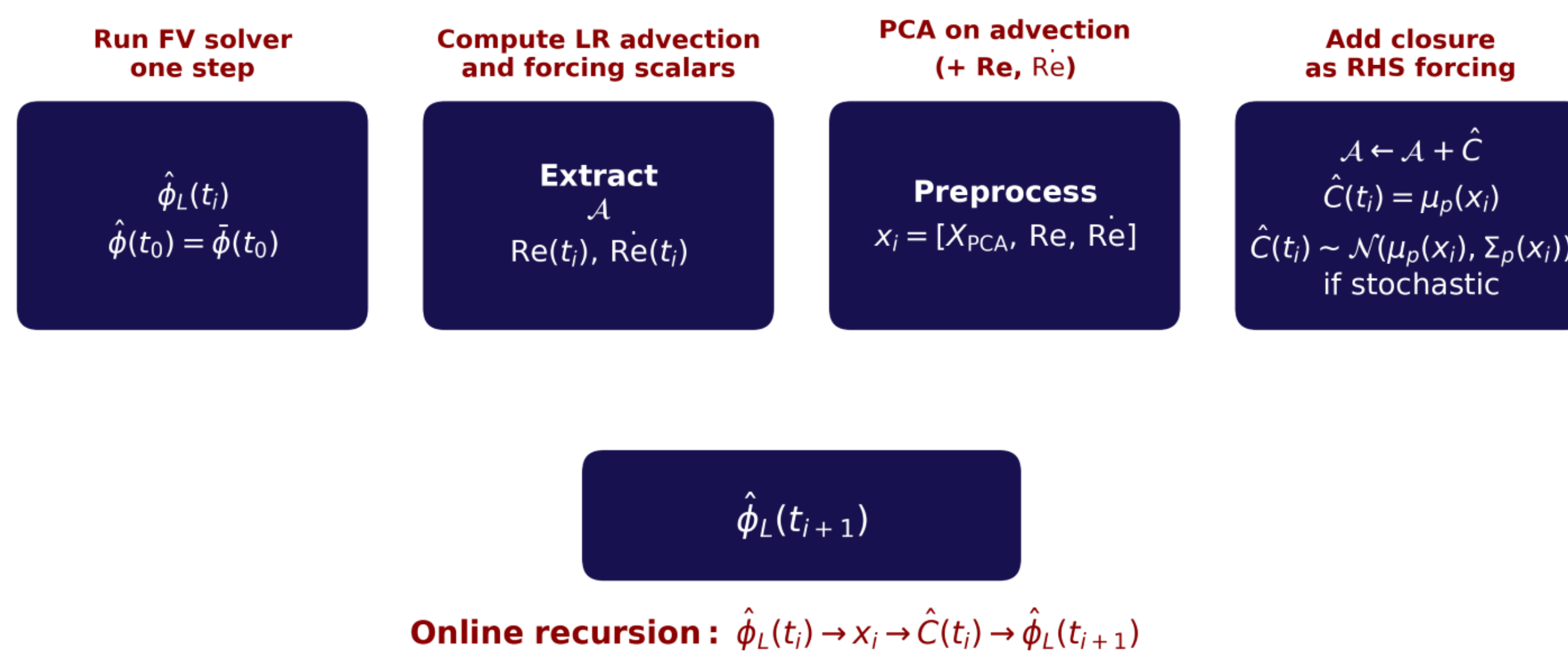
Linear Model of coregionalization (LMC)

$$g_q(x) \sim \mathcal{GP}(0, k_q(x, x')), \quad q = 1, \dots, Q$$

$$f_p(x) = \sum_{q=1}^Q a_{pq} g_q(x)$$

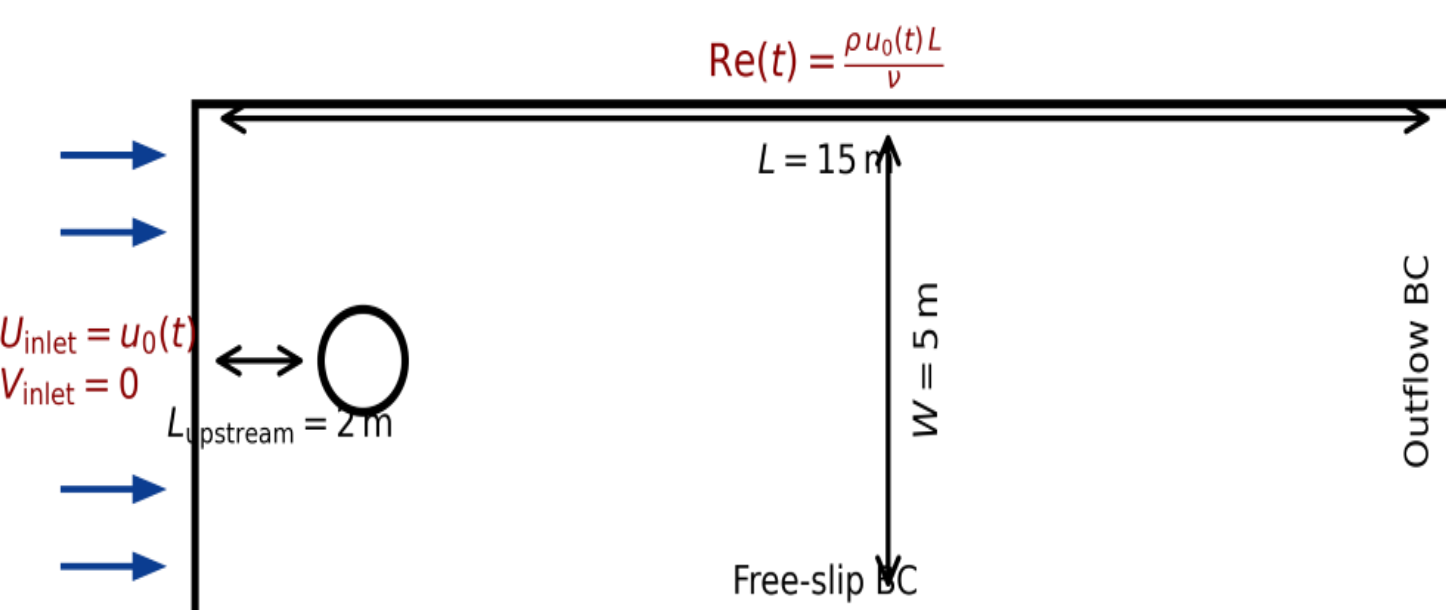
$$K_f(x, x') = A K_g(x, x') A^\top$$

SVGP Inference: Online Finite volume-GP predictions

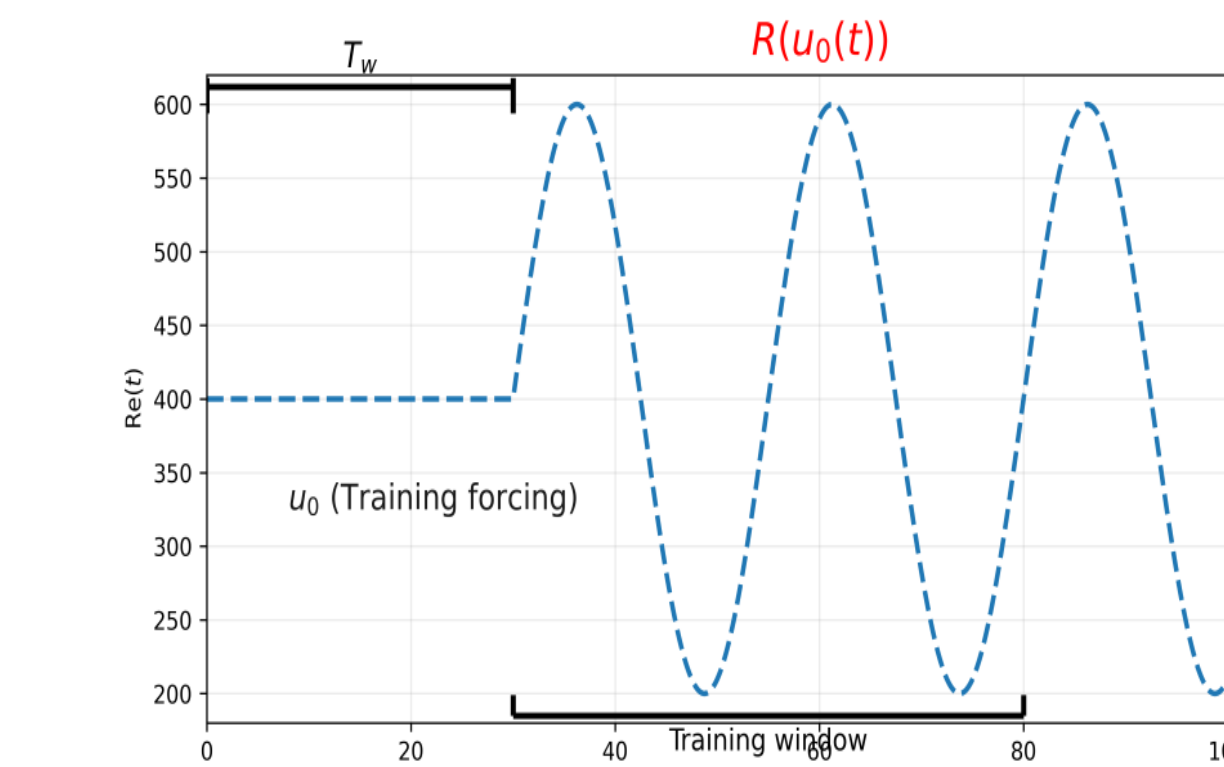


Simulation Training data and Training configuration

Simulation Setup



Training and Prediction Setup



- Resolved grid:** $(N_x, N_y) = 1200 \times 400$
- Under-resolved grid:** $(N_x, N_y) = 150 \times 50$
- Shedding Period:** $T_{\text{shedding}} = 5\text{ s}$
- Re modulation Period:** $T = 5 * T_{\text{shedding}}$
- Training window:** $T_{\text{train}} = 2 * T = 50$
- # Eddy shedding:** 10
- Re Training range:** 200-600

- Training samples correspond to **2 Re modulation cycles** which corresponds to **ten eddies (events)**.
- SVGP model is only trained **once** on simulation with Re modulation function $u_0(t)$ and in range Re
- In distribution** samples are flow fields modulated with $u_0(t)$ and in range Re
- Out of distribution** samples are flow fields modulated with $u(t) \neq u_0(t)$ and/or out-of-range Re

Application #1 : In distribution inference: Periodically Modulated Flow

Past an Obstacle

Experiment goals

Physics objectives

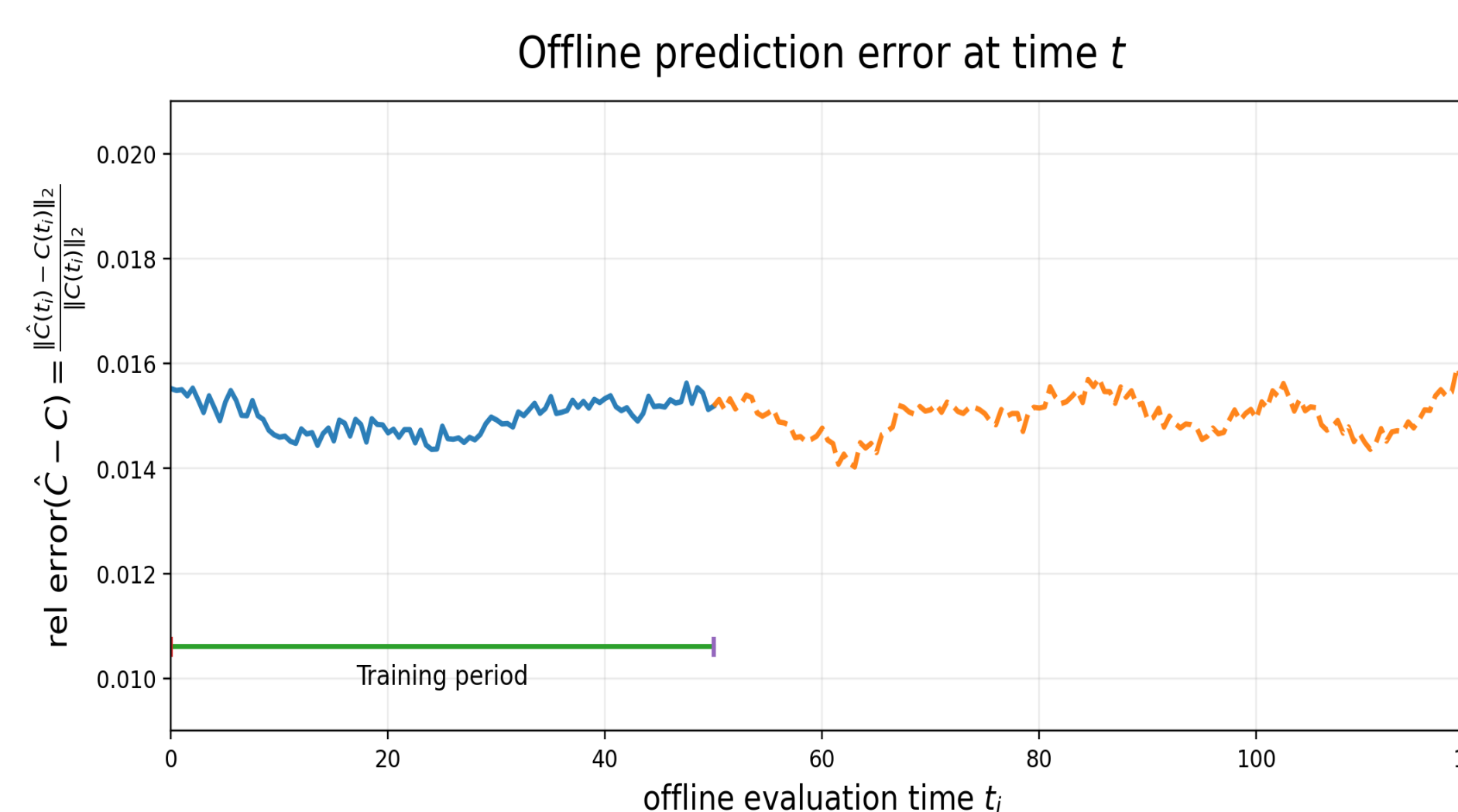
- Learn transient wake dynamics under modulated $\text{Re}(t)$
- Capture vortex shedding frequency shifts
- Represent acceleration via $d\text{Re}/dt$

Statistical learning

- Learn time dependent closure statistics
- Maintain accuracy under slow distribution drift
- Generalize beyond training cycles

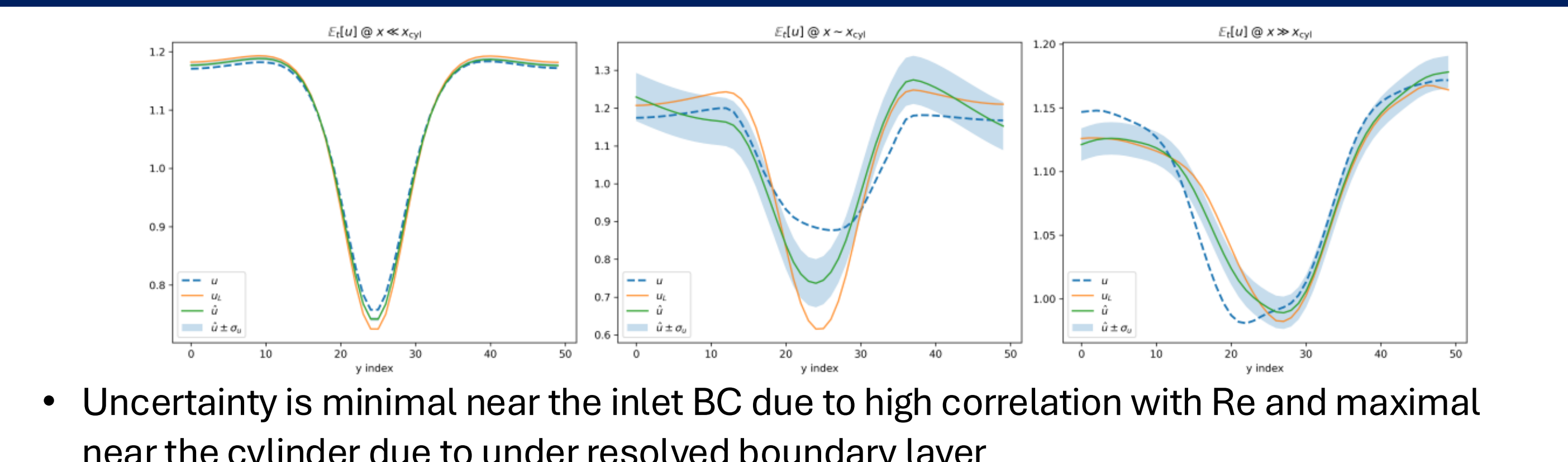
Online deployment

- Stable FV-GP coupling
- Bounded long horizon growth
- Detect OOD via predictive variance
- Robustness under recursive feedback

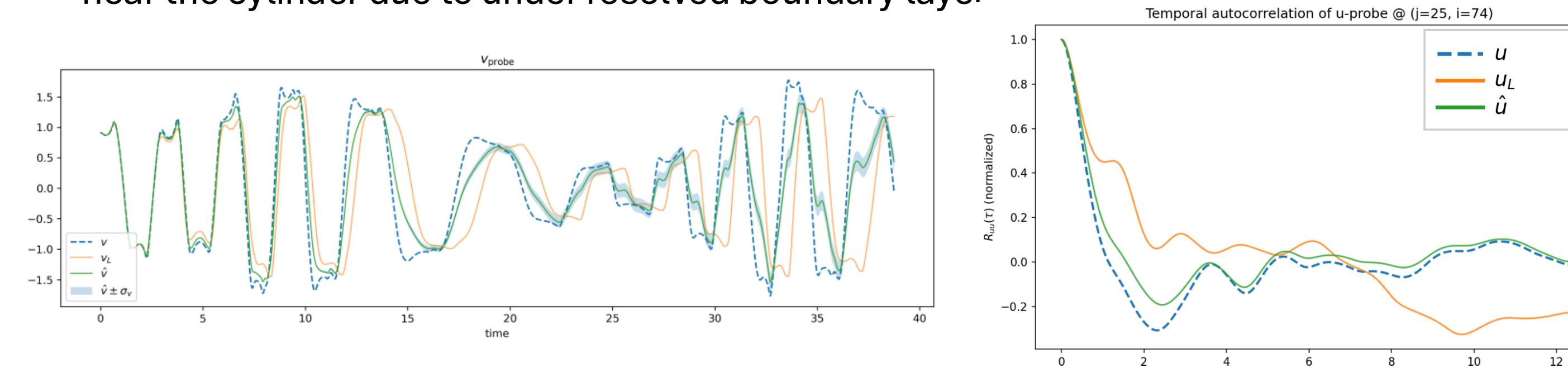


- Offline closure evaluations are successful with little to no dependence on the time of the evaluation. Relative error varies between **1.4 and 1.6%**

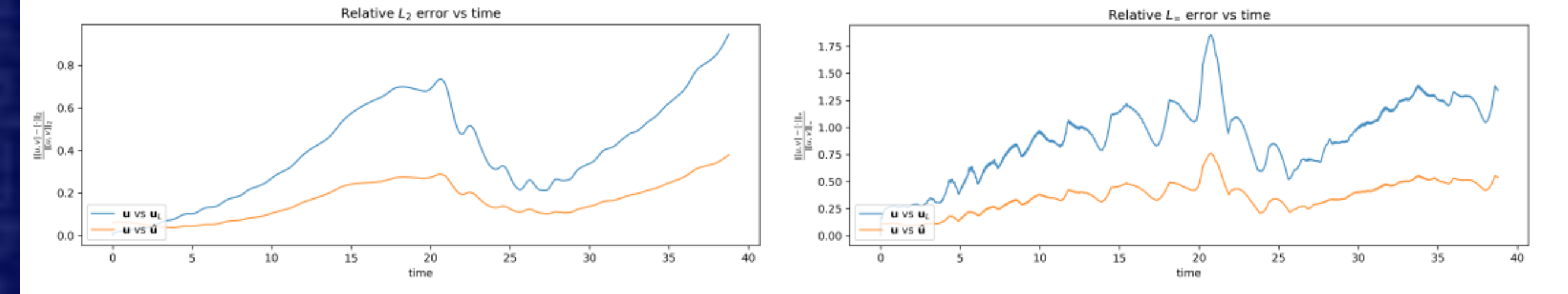
Application #1 : Inference results and discussions



- Uncertainty is minimal near the inlet BC due to high correlation with Re and maximal near the cylinder due to under resolved boundary layer



Application #1 : Inference results and discussions



- Error accumulation causes distribution drift and uncertainty grows over time
- Online predictions are stable with bounded error growth

Application #2 : Out of distribution inference configuration

Experiment goals

- All the goals from Application #1
- Online FV-GP inference on out of distribution samples where Re modulation forcing are **different from Training** or Reynolds number is **outside of Training Re range**
- Quantify epistemic uncertainty

Vorticity difference: In-vs Out-of-distribution

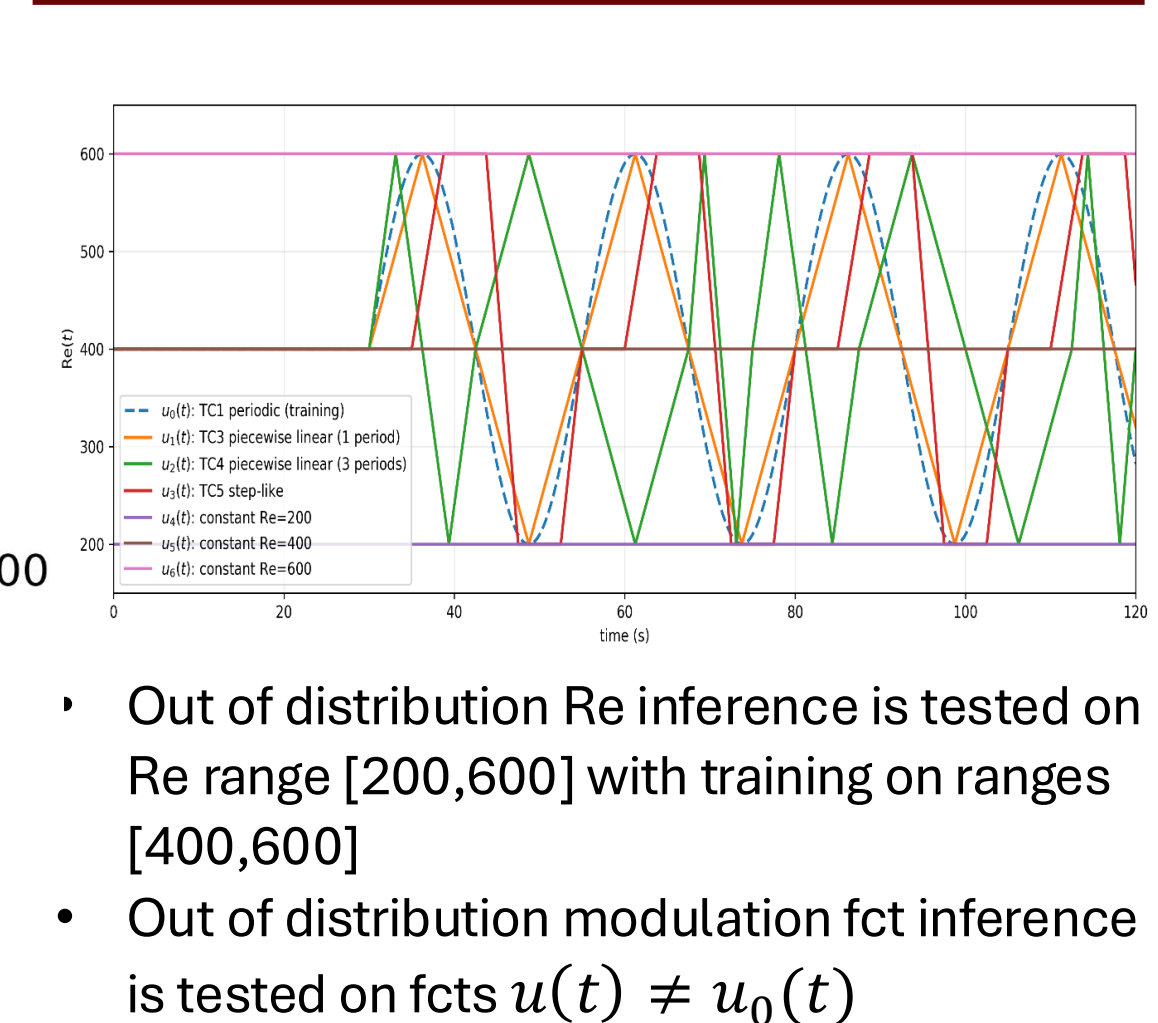
Out of distribution Re (TC2) @ $t = 61.25\text{ s}$, $\text{Re} = 600$



Out of distribution modulation fct (TC4), @ $t = 71.25\text{ s}$, $\text{Re} = 600$



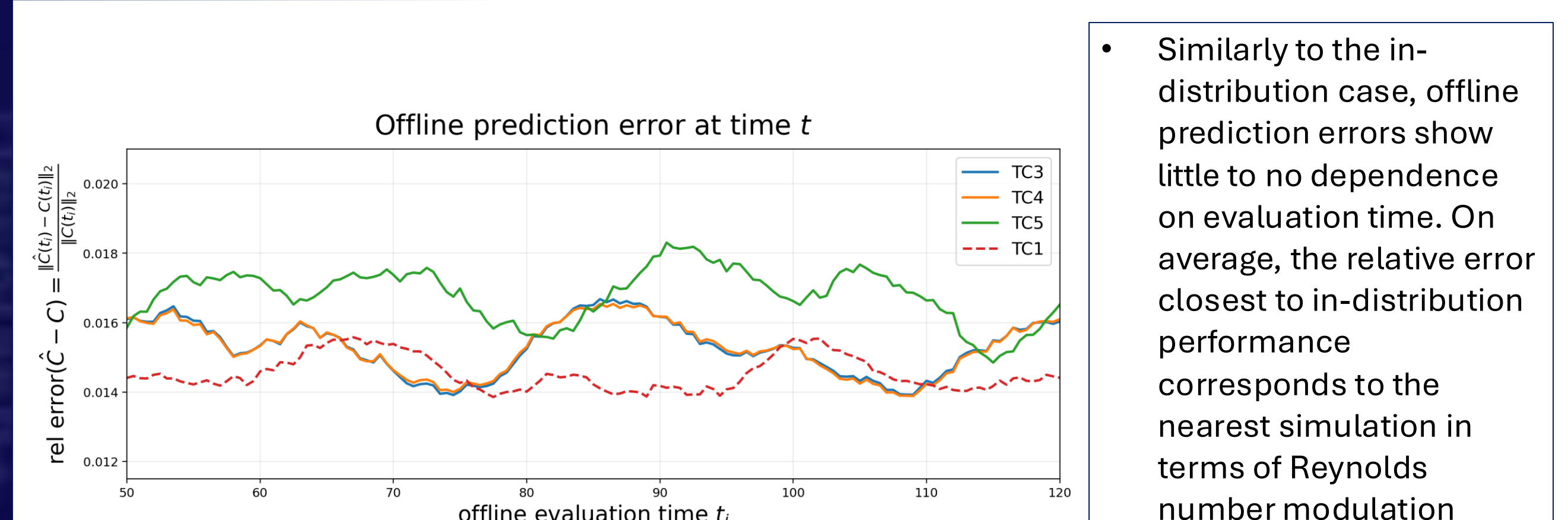
OOD Re modulation functions



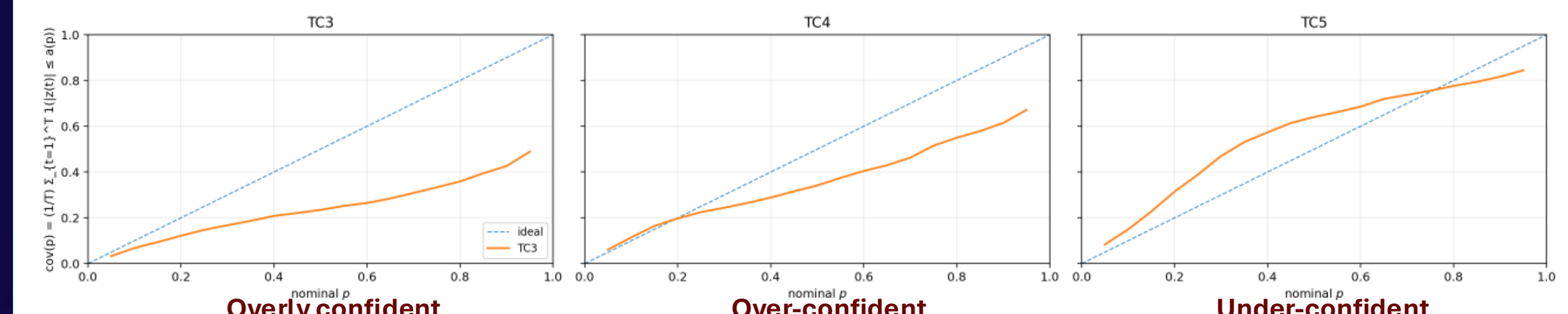
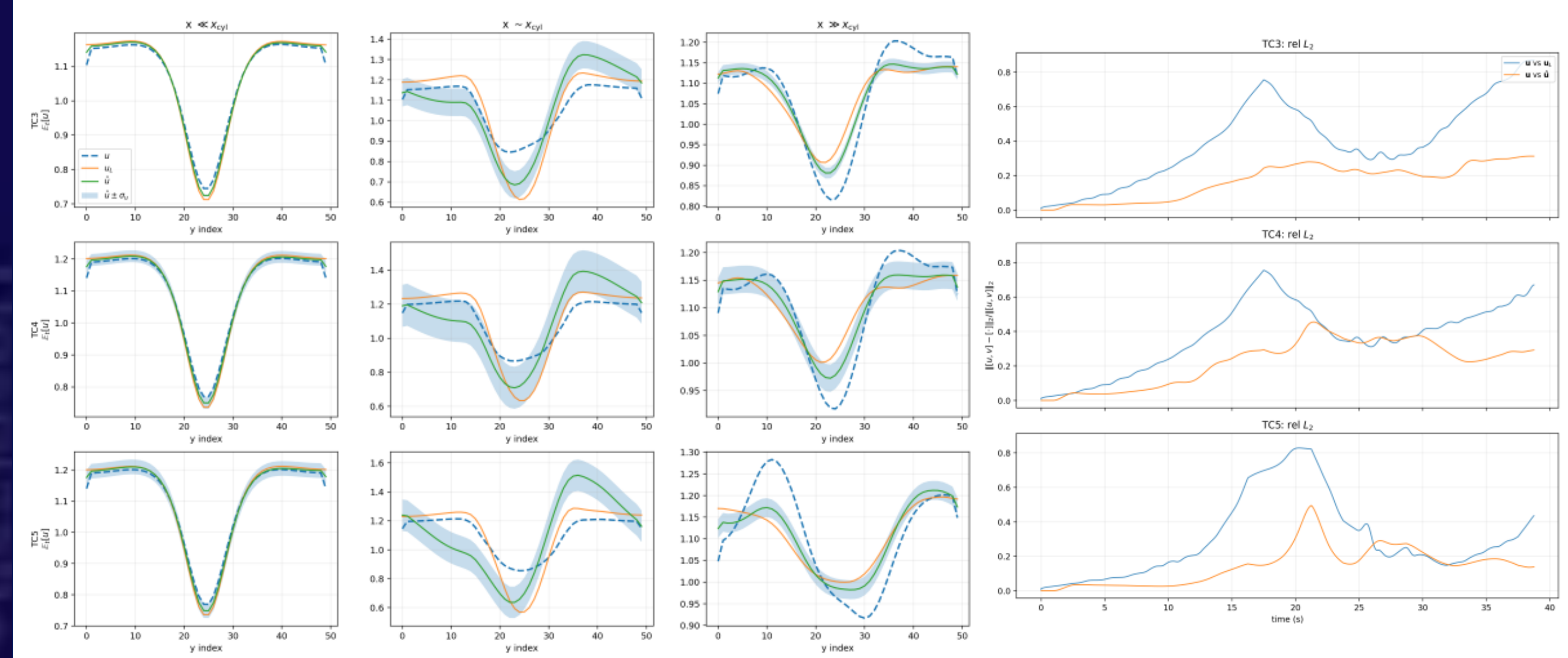
Test case	Training forcing	Testing forcing	Training Re range	Testing Re range
TC1	u_0	u_0	[200, 600]	[200, 600]
TC2	u_0	u_0	[400, 600]	[200, 600]
TC3	u_0	u_1	[200, 600]	[200, 600]
TC4	u_0	u_2	[200, 600]	[200, 600]
TC5	u_0	u_3	[200, 600]	[200, 600]
TC6	u_0	u_4	[200, 600]	[200]
TC7	u_0	u_5	[200, 600]	[400]
TC8	u_0	u_6	[200, 600]	[600]

- Results below are shown for **TC3, TC4** and **TC5**. With the note that **TC6-TC8** had the smallest error.

Application #2 : Inference results and discussions



- Similarly to the in-distribution case, offline prediction errors show little to no dependence on evaluation time. On average, the relative error closest to in-distribution performance corresponds to the nearest simulation in terms of Reynolds number modulation

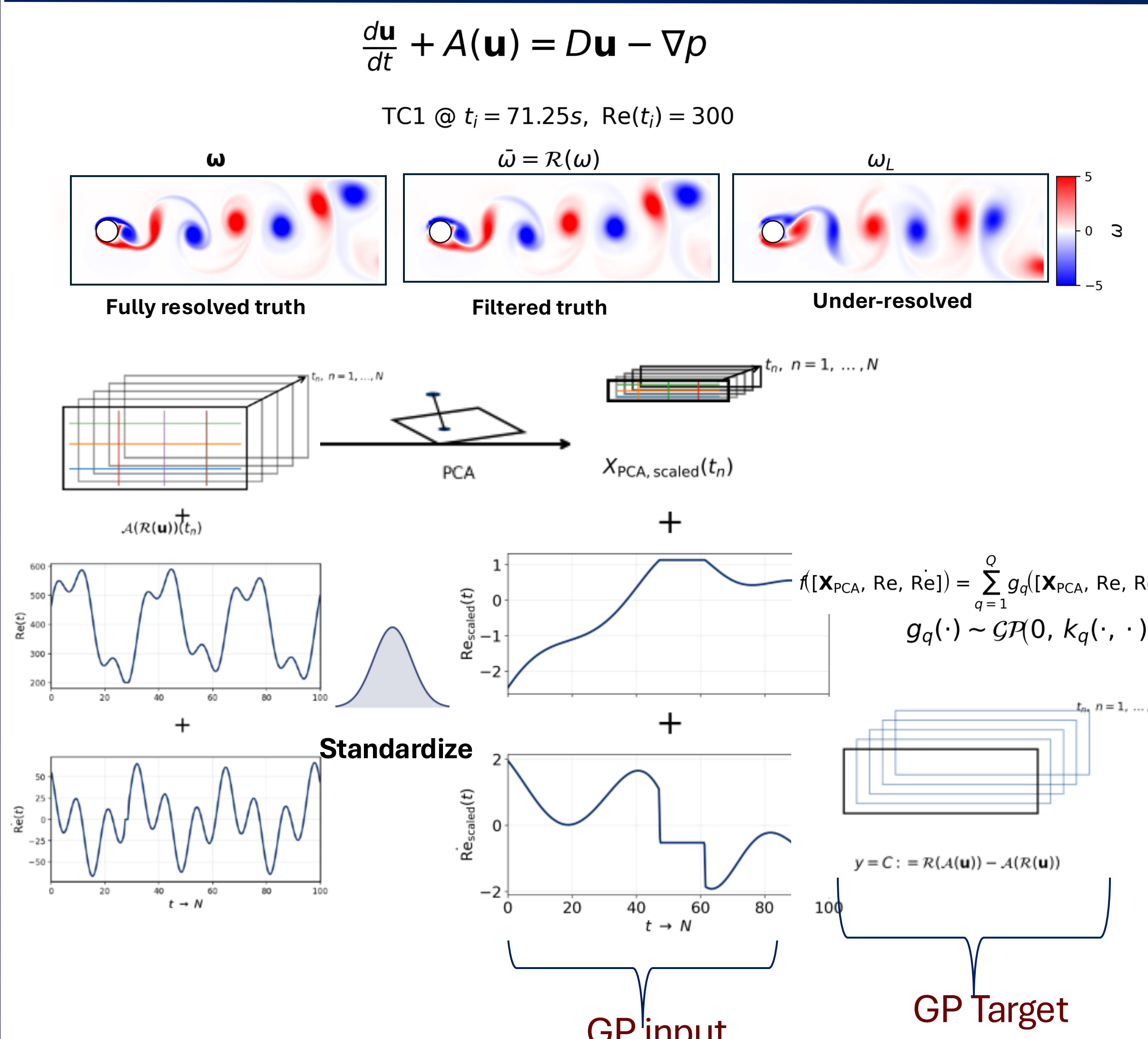


- Uncertainty has similar properties to the in-distribution prediction case and is overall higher
- Under/over-dispersion (over/under-confidence) is governed by $d\text{Re}/dt$. GP is over-confident when $d\text{Re}/dt$ lies close to training support.
- Spatial autocorrelations are still very well captured under OOD Re modulation functions
- Online predictions are stable with bounded error growth
- Inference was **not successful** on OOD cases with **Out Of distribution Re ranges** (TC2)

Conclusion

We developed and trained sparse variational Gaussian Process (GP) closure models for a Finite Volume framework that seamlessly inject mean or stochastic GP closure terms inside coarse simulations and tested them on in- and out-of-distribution flows. **Acknowledgments** We are grateful to the Office of Naval Research for partial support under grant N00014-20-1-2023 (MURI ML-SCOPE) to the Massachusetts Institute of Technology

SVGP Training: Closure definition, Inputs and Outputs



Variable	Definition	Range	Space
n	data index	$1, \dots, N$	—
N	number of training points	$N \in \mathbb{N}$	—
d	input dimension	$d \in \mathbb{N}$	$x_n \in \mathbb{R}^d$
D	number of outputs / tasks	$D \in \mathbb{N}$	$y_n \in \mathbb{R}^D$
Q	number of latent GPs (LMC)	$1, \dots, Q$	$g_q: \mathbb{R}^d \rightarrow \mathbb{R}$
M	number of inducing points	$1, \dots, M$	$Z \in \mathbb{R}^{M \times d}$
A	mixing (coregionalization) matrix	$A \in \mathbb{R}^{D \times Q}$	$f(x) = A g(x)$

SVGP Training: Kernel definition and learning objectives

$$m_q(x) = b_q + a_q \text{Re}_{\text{scaled}}$$

$$k_q(x, x') = k_q, \text{RBF}(x_k, x'_k) + k_q, \text{Matérn}(3/2)(x_k, x'_k), \quad x_k = [X_{\text{PCA, scaled}}, \text{Re}_{\text{scaled}}]$$

$$\mathcal{J} = -\text{ELBO} + \beta_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{Re}} \|\theta_{\text{mean}}\|_2^2 + \mathcal{L}_{\text{WNLL}}(a_w) + \mathcal{L}_{\text{L, cap}}(\lambda_t, \log \ell_{\text{max}})$$

Where:

$$\|\theta\|_2^2 = \sum_{q=1}^Q a_q^2 + \sum_{q=1}^Q b_q^2$$

$$\mathcal{L}_{\text{bal}} = (\mathcal{E}(\Omega_{\text{wake}}) - \mathcal{E}(\Omega_{\text{wake}}^c))^2$$

$$\mathcal{L}_{\text{L, cap}} = \lambda_t E_q[d] \max(0, \log \ell_q, d - \log \ell_{\text{max}})^2$$

$$\mathcal{L}_{\text{WNLL}} = \frac{1}{N} \sum_{p,n} w_{p,n} \left[\frac{1}{2} \log \sigma_{p,n}^2 + \frac{(y_{p,n} - \mu_{p,n})^2}{2\sigma_{p,n}^2} \right]$$